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# Residual stresses prediction in machining of Inconel 718 superalloy using a constitutive model considering the state of stress

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## Abstract

The need for developing new products that meet high functional requirements, such as high fatigue strength, superior thermal and corrosion resistance, have permitted to focus on the development of enhanced materials and seek solutions that also promote better surface integrity. It is the case of the Inconel 718 nickel superalloy, where the combination of its mechanical and thermal properties provides a full industrial application, especially in hot engine components in the aerospace industry. However, there is still a big challenge to obtain suitable surface integrity for this kind of application after machining this alloy. This work aims to predict the surface integrity in the machining of the Inconel 718 alloy using an orthogonal cutting model implemented in Abaqus FEA software. This model includes a constitutive model considering the most relevant factors affecting the mechanical behavior of Inconel 718 in machining, including the strain rate and the state of stress. The machining model is validated by comparing the measured and predicted results. Then, it is used to evaluate the influence of the cutting regime parameters and tool geometry on the residual stresses.

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**Keywords:** Surface Integrity; Modelling; Inconel 718.

## 1. Introduction

Inconel 718 is a nickel-based superalloy widely used in high-performance applications, especially those requiring excellent material properties for working at high temperatures. Widely used in the design and manufacture of hot turbine parts in for the aerospace industry, this alloy is a high-strength and corrosion-resistant nickel-based superalloy, with a high content of Ni, Cr, and Fe [1–4].

It should be noted that due to its high strain-hardening tendency, low thermal conductivity, Inconel 718 is characterized as a difficult-to-cut material. The optimization of the cutting process of this material is very costly by using the traditional experimental approach. Therefore, applying a numerical simulation approach will help to minimize production costs and time significantly [5–7].

Several numerical models are available in the literature for predicting the surface integrity in the machining of Inconel 718. Arrazola et al. [8] proposed a 3D thermo-mechanically model of face turning of Inconel 718 (45 HRC) alloy heat-treated at 955°C and subsequently quenched for one hour. This model is based on the updated incremental Lagrangian formulation. They use a modified J-C constitutive model proposed by Ulutan et al. [9]. Accounting for a temperature-dependent flow (strain) softening effect, in addition to strain hardening and strain rate and thermal softening. They found a non-negligible difference between predicted and measured in-depth residual stress profiles. This difference is significant for depths greater than 25 µm. Jafarian et al. [10] proposed a 2D thermo-mechanical model of the orthogonal cutting process of aged Inconel 718 under plane strain conditions. An update-Lagrangian formulation with the remeshing technique was

chosen to simulate the cutting process. Cockroft and Latham's damage model was used to simulate the chip segmentation. They evaluated the influence of several constitutive models with different sets of coefficients available in the literature on the microhardness and grain size in the surface and subsurface of the workpiece. The constitutive models used in their work are the Johnson-Cook (J-C) [11], and modified versions of this model to intensify the effects of strain hardening, strain rate and temperature in the mechanical behavior of the Inconel 718. They found a difference between predicted and measured results (forces, chip geometry, and cutting temperature) between 8 and 50%, depending on the model. They concluded that the constitutive model has a significant effect on the results of the simulation, and a suitable selection of the model and corresponding coefficients is essential for an accurate simulation of the cutting process.

### Nomenclature

|                             |                                                                  |
|-----------------------------|------------------------------------------------------------------|
| $C_0$                       | yield stress                                                     |
| $C_1$                       | material coefficients in Plastic Strain Hardening (PSH) law      |
| $C_2$ and $C_4$             | material PSH exponents                                           |
| $C_3$                       | material coefficients in PSH law                                 |
| $c_1, c_2$ and $c_3$        | material coefficients in Modified Mohr-Coulomb (MMC) model       |
| $c_\eta$                    | coefficient of stress triaxiality dependency on plasticity       |
| $c_\theta^{ax}, c_\theta^s$ | material parameters in plasticity model by Bai and Wierzbicki    |
| $D_{sr}$                    | strain rate dependency coefficient in plastic model              |
| $D_1$                       | strain-rate dependency constant in the model of Mohr-Coulomb MMC |
| $D_2$                       | temperature dependency constant in MMC                           |
| $D_c$                       | critical damage indicator                                        |
| $\beta$                     | flow stress softening coefficient                                |
| $\theta$                    | Lode angle                                                       |
| $\bar{\theta}$              | Lode angle parameter                                             |
| $\gamma$                    | Lode parameter                                                   |
| $\eta_0$                    | stress triaxiality reference                                     |
| $\dot{\epsilon}_0$          | reference strain-rate                                            |
| $k$                         | material coefficient inserted by Algarni                         |
| $m$                         | thermal softening coefficient                                    |
| $m_f$                       | post-failure softening power coefficient                         |
| $N$                         | material PSH exponent                                            |
| $T_{melt}$                  | melting temperature                                              |
| $T_{room}$                  | room temperature                                                 |

In order to accurately predict the residual stresses in machining Inconel 718, a suitable constitutive model is required. This model should accurately reproduce the mechanical behavior of Inconel 718 in machining under a different state of stresses [1,2]. The state of stress is represented by the stress triaxiality ( $\eta$ ) and the Lode angle ( $\theta$ ) parameter, cf. [12]. Algarni et al [1] have proposed a constitutive model for the Inconel 718 including the effects of stress triaxiality ( $\eta$ ) and Lode angle ( $\theta$ ) on both material plasticity and fracture. However, the model used by these authors does not consider

the effects of strain rate and thermal softening on mechanical behavior, both important for machining simulation.

Moreover, the failure model was calibrated by considering tensile specimens only (positive stress triaxiality), thus not including the effect of the negative triaxiality on fracture, often observed in metal cutting. Sjöberg et al. [2] evaluated the accuracy of three failure models to describe the fracture of Inconel 718. They concluded that the Modified Mohr-Coulomb (MMC) model presents the best result for the Inconel 718.

This work aims to propose an orthogonal cutting model of Inconel 718, which includes a modified Algarni and Sjöberg plasticity model [1,2], to predict the thermomechanical phenomena and the residual stresses. Moreover, the Mohr-Coulomb model modified by Sjöberg [2] will be used to describe damage initiation and evolution.

## 2. Constitutive model for Inconel 718 considering the state of stress

Based on Algarni et al. [1] and Johnson-Cook [11] plasticity models, a material plasticity model is proposed to describe the mechanical behavior of Inconel 718 in the cutting process. It considers five effects on material plasticity: stress triaxiality, Lode angle, strain hardening, thermal softening, and strain rate. The proposed plasticity model is given by the following equation:

$$\sigma(\bar{\epsilon}_{pl}, \eta, \theta, \dot{\epsilon}, T) = ((C_0 + C_1 \cdot (\bar{\epsilon}_{pl})^{C_2} - C_3 \cdot e^{C_4 \cdot \bar{\epsilon}_{pl}})) \cdot (1 - c_\eta \cdot (\eta - \eta_0)) \cdot \left( c_\theta^s + (c_\theta^{ax} - c_\theta^s) \left( \frac{k+1}{k} \right) \left( \gamma - \frac{\gamma^{k+1}}{k+1} \right) \right) \cdot \left( 1 + D_{sr} \cdot \ln \left( \frac{\dot{\epsilon}}{\dot{\epsilon}_0} \right) \right) \cdot \left( 1 - \left( \frac{T - T_{room}}{T_{melt} - T_{room}} \right)^m \right) \quad (1)$$

To simulate chip formation and chip segmentation, a damage model is used. Sjöberg et al. [2] model is used to describe the damage initiation (Eq. 2), while Li and Wierzbicki [13] model is used to describe damage propagation (Eq. 3). In the last case, the stiffness material will be fully degraded when the damage,  $D$ , is greater than the critical damage indicator,  $D_c$ .

$$\epsilon_f(\eta, \bar{\theta}, \dot{\epsilon}, T) = \left\{ c_2 \times \left[ c_3 + \frac{\sqrt{3}}{2 - \sqrt{3}} (c^{ax} - c_3) \left( \sec \left( \frac{\bar{\theta}\pi}{6} \right) - 1 \right) \right] \times \right\}^{\frac{-1}{N}} \cdot \left( 1 + \left[ \left( \frac{1+c_1^2}{3} \right)^{0.5} \cos \left( \frac{\bar{\theta}\pi}{6} \right) + c_1 \left( \eta + \frac{1}{3} \sin \left( \frac{\bar{\theta}\pi}{6} \right) \right) \right] \right)^{D_1} \cdot (1 + D_2 T^*) \quad (2)$$

$$\tilde{\sigma} = \beta \cdot \sigma(\bar{\epsilon}_{pl}, \eta, \theta, \dot{\epsilon}) \begin{cases} D < 1 \therefore \tilde{\sigma} = \sigma(\bar{\epsilon}_{pl}, \eta, \theta, \dot{\epsilon}) \\ D = 1 \therefore \beta = 1 \text{ (damage threshold)} \\ D \geq 1 < D_c \therefore 0 < \beta \leq 1 \\ D \geq D_c \therefore \beta = 0 \end{cases} \quad (3)$$

where

$$\beta = \left( \frac{D_c - D}{D_c - 1} \right)^{m_f} \quad (4)$$

$mf$  and  $Dc$  coefficients were chosen based on the condition of material instability in an element in the post-initiation state according to the material failure patterns [1,13].

The coefficients of both plasticity and damage models were extracted from Sjöberg [2], and Algarni [1] work and their values are gathered in Table 1.

Table 1. Constitutive models coefficients for plasticity and damage [1,2].

| $C_0$           | $C_1$ | $C_2$    | $C_3$              | $C_4$ | $C_\eta$   | $\eta_0$   | $c_\theta^s$ |
|-----------------|-------|----------|--------------------|-------|------------|------------|--------------|
| MPa             | MPa   | -        | MPa                | -     | -          | -          | -            |
| 1330            | 704   | 0.25     | 342                | 9.79  | 0.11       | 0.333      | 1            |
| $c_\theta^{ax}$ | k     | $D_{sr}$ | $\dot{\epsilon}_0$ | m     | $T_{room}$ | $T_{melt}$ |              |
| -               | -     | -        | 1/s                | -     | °C         | °C         |              |
| 1               | 0.75  | 0.00355  | 1                  | 1.75  | 20         | 1500       |              |
| $c_1$           | $c_2$ | $c_3$    | $c^{ax}$           | N     | $D_1$      | $D_2$      |              |
| -               | -     | -        | -                  | -     | -          | -          |              |
| 0.1             | 2.01  | 0.931    | 1                  | 0.127 | -0.00403   | -0.255     |              |

### 3. Numerical model and parameters

The proposed 2D orthogonal cutting model is detailed in Fig. 1. This model, as implemented in ABAQUS/Explicit using the Lagrangian formulation. The workpiece has meshed with 21000 elements and its mechanical behavior is represented by the proposed constitutive model described in the previous section (Eq. 1, 2, and 3). This model was implemented in ABAQUS through a VUMAT user subroutine. The chip formation is simulated using the element deletion technique. The tool geometry and its thermo-mechanical properties were extracted from Outeiro et al. [14]. It is an uncoated tungsten carbide tool, having a cutting edge radius,  $r_n$ , of 30  $\mu\text{m}$ , a flank angle,  $\alpha$ , of 7°, and three values of the rake angles,  $\gamma$ , (-6°, 6° and 10°) are used. The tool is defined as deformable and meshed with 399 elements. The contact model between the workpiece and the cutting tool is represented by Zorev's model [15]. The friction coefficient is given in function of the sliding speed, based on the works of Zemzemi et al. [16] and Mondelin et al. [17].

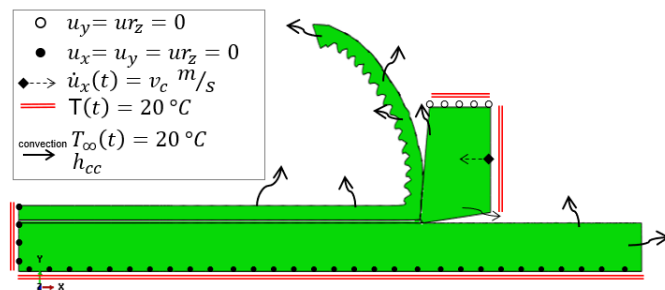


Fig. 1. Geometry, meshing, and boundary conditions of the FE model.

Regarding the boundary conditions, the model is pinned at the bottom and left surfaces of the workpiece. The cutting speed is applied to the top surface of the cutting tool in the x-direction, while no motion is imposed in the y-direction. Thermal dissipation by forced convection with the air is applied in the upper and right surfaces of the workpiece, and the bottom

and left surfaces of the tool, using a convection coefficient of 20 W/m<sup>2</sup>.K. Constant temperature of 20° is applied to the bottom and left surfaces of the workpiece, and upper and right surfaces of the tool.

As far as the cutting regime parameters are concerned, three values of the uncut chip thickness,  $h$  (0.05 mm, 0.15 mm and 0.25 mm), and three values of cutting speed,  $v_c$  (30 m/min, 55 m/min and 90 m/min) were used. The width of cut,  $w$ , is constant and equal to 4 mm. Table 2 resumes the cutting conditions used in the simulations and experimentally from the work of Outeiro et al. [14].

Table 2. Cutting conditions for simulations (SIM) and experiments (EXP) (constant edge radius of 30  $\mu\text{m}$ , flank angle of 7°, and width of cut of 4 mm).

| Cut. Cond. ID | Used in simulation and/or experimentally | Uncut chip thickness (mm) | Cutting speed (m/min) | Rake angle (°) |
|---------------|------------------------------------------|---------------------------|-----------------------|----------------|
| S1            | SIM                                      | 0.05                      | 30                    | -6             |
| S2            | SIM                                      | 0.05                      | 90                    | -6             |
| S3            | SIM                                      | 0.25                      | 90                    | -6             |
| S4            | SIM                                      | 0.25                      | 90                    | 10             |
| S5            | SIM&EXP                                  | 0.15                      | 55                    | 6              |
| S6            | SIM&EXP                                  | 0.15                      | 90                    | 6              |

### 4. Results and discussion

#### 4.1. Comparison between measured and predicted results

In order to evaluate the ability of the proposed orthogonal cutting model to predict the thermomechanical phenomena and the residual stresses in the orthogonal cutting simulation of Inconel 718, results obtained from the numerical simulations are compared with those obtained experimentally from the work of Outeiro et al. [14]. Two cutting conditions were selected for this comparison, as shown in Table 2 (S5 and S6).

Fig. 2 shows both predicted and measured cutting forces for the two cutting conditions.

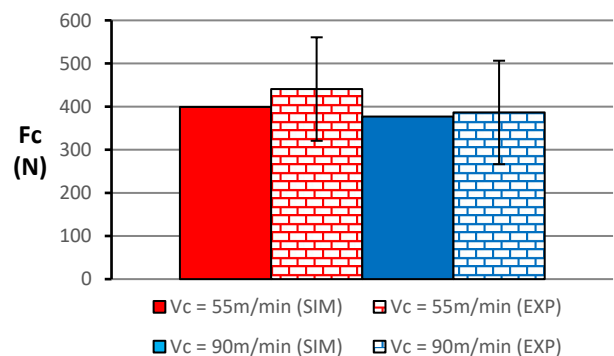


Fig. 2. Measured and simulated cutting forces ( $h = 0.15$  mm,  $w = 4$  mm,  $\alpha = 7^\circ$ ,  $\gamma = 6^\circ$ ,  $r_n = 30$   $\mu\text{m}$ ).

The difference between measured and predicted cutting force is very small (less than 10% for the lowest cutting speed) and within the experimental error bar. This error is event lower for the highest cutting speed.

Fig. 3 shows the maximum temperature at the tool-chip interface, the called cutting temperature,  $T_c$ . As shown by this figure, the simulated values are quite close to the experimental ones.

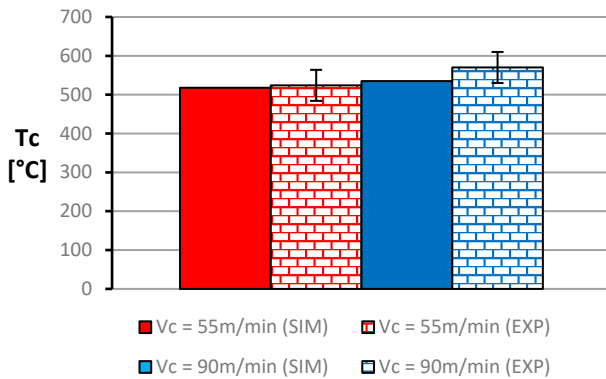


Fig. 3. Measured and simulated cutting temperatures ( $h = 0.15$  mm,  $w = 4$  mm,  $\alpha = 7^\circ$ ,  $\gamma_o = 6^\circ$ ,  $r_n = 30$   $\mu$ m).

Fig. 4 shows both measured and simulated chip geometry parameters: peak, valley, pitch. This figure shows that the model underestimates the chip peak and slightly overestimates the chip valley and pitch. However, the predicted chip geometry parameters are within the experimental error bar.

Chip Compression Ratio (CCR) is the ratio between the chip thickness and the uncut chip thickness. Since the chip thickness is not constant, CCR will also change from a minimum value corresponding to the minimum chip thickness (chip valley), to a maximum value corresponding to the maximum chip thickness (chip peak). Predicted and measured CCR are shown in Fig. 5 for two cutting speeds. This figure shows that the model slightly underestimates this parameter, although the simulated CCR is almost within the experimental error bar.

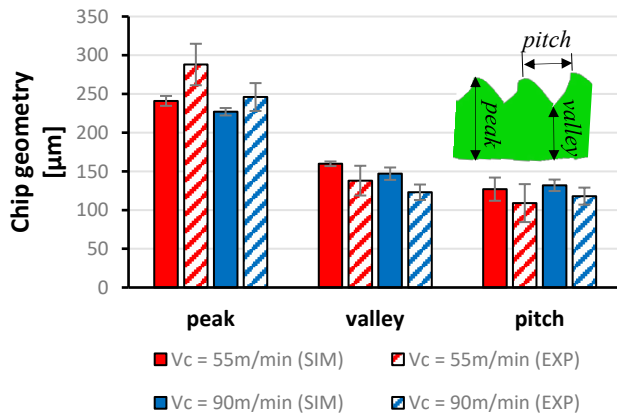


Fig. 4. Measured and simulated chip geometry parameters ( $h = 0.15$  mm,  $w = 4$  mm,  $\alpha = 7^\circ$ ,  $\gamma_o = 6^\circ$ ,  $r_n = 30$   $\mu$ m).

To calculate the residual stresses, the distributions of temperature, stress, and strain in the machined workpiece are transferred to another model. This model consists into unloading and cooling the workpiece until reaching the mechanical equilibrium and room temperature in the workpiece. This model is simulated in ABAQUS/Standard using the proposed constitutive model in a UMAT user subroutine. After residual stress calculation, a procedure

proposed by Outeiro [14,18] to extract them from the FEM model is applied.

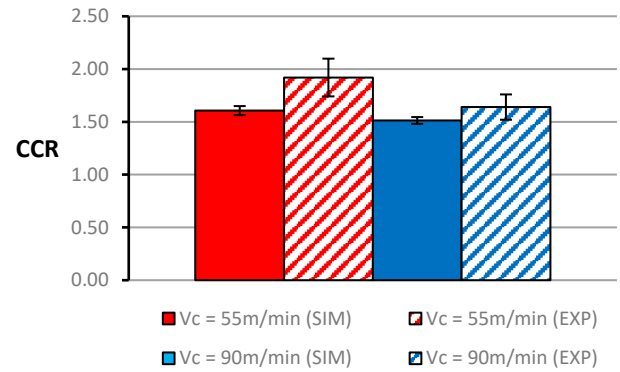


Fig. 5. Measured and simulated chip compression ratio (CCR) ( $h = 0.15$  mm,  $w = 4$  mm,  $\alpha = 7^\circ$ ,  $\gamma_o = 6^\circ$ ,  $r_n = 30$   $\mu$ m).

Fig. 6 shows both residual stresses in the cutting direction ( $\sigma_{//}$ ) and the direction normal to the cutting direction ( $\sigma_{\perp}$ ) in the function of the distance from the machined surface for a cutting speed of 55 m/min. It is noteworthy that the simulated results are presented with error bars since, due to the cyclic nature of chip formation, the residual stresses also vary cyclically along the cutting direction [8,19,20], as can be seen in Fig. 7. Consequently, high residual stresses variation is obtained near the machined surface. This variation decreases as the distance from the machined surface increases (Fig. 6).

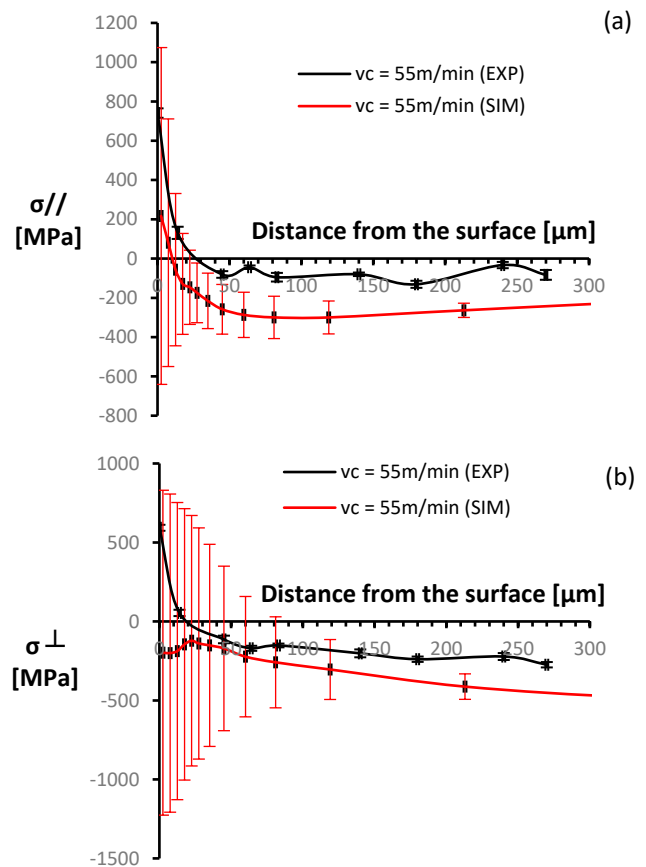


Fig. 6. Measured and simulated residual stresses in the (a) cutting direction (b) and normal to these directions ( $h = 0.15$  mm,  $w = 4$  mm,  $\alpha = 7^\circ$ ,  $\gamma_o = 6^\circ$ ,  $r_n = 30$   $\mu$ m).

In general, both  $\sigma_{//}$  and  $\sigma_{\perp}$  are reasonably predicted, considering the variation of the simulated residual stresses in the function of the distance from the machined surface.

In general, the developed orthogonal cutting model that includes a new proposed constitutive model can represent the cutting process of Inconel 718 with enough accuracy. Therefore, this model can be applied to investigate the influence of the cutting conditions on the residual stresses in the machined surface and subsurface of Inconel 718.

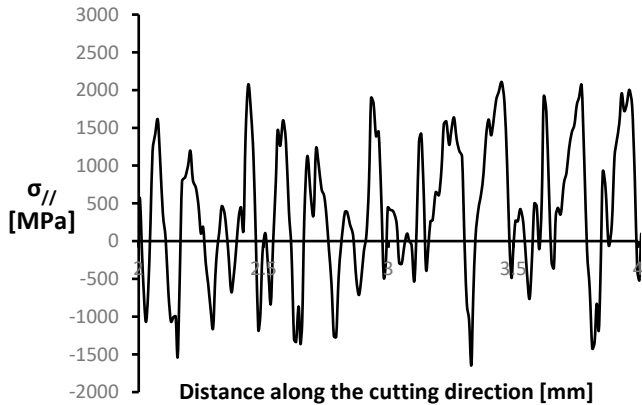


Fig. 7. Variation of the circumferential residual stresses on the machined surface ( $v_c = 55$  m/min,  $h = 0.15$  mm,  $w = 4$  mm,  $\alpha = 7^\circ$ ,  $\gamma_o = 6^\circ$ ,  $r_n = 30$   $\mu$ m).

#### 4.2. Influence of the cutting parameters on the residual stresses

Since the residual stresses in the cutting direction ( $\sigma_{//}$ ) are higher than those in the other direction, so the most critical for the functional performance and life of the components, only this stress ( $\sigma_{//}$ ) will be presented in this section.

##### 4.2.1. Approach to explain the influence of the cutting parameters on the residual stresses

To explain the influence of cutting parameters in the residual stresses, the model proposed by Wu and Matsumoto [22] and Denguir et al. [21] will be used. Wu and Matsumoto [22] proposed a simple approach to describe the residual stress formation in metal cutting due to the mechanical action (thermal effects are neglected) of the cutting tool over the workpiece, by using a stress-strain curve obtained in a standard uniaxial tensile test. Denguir et al. [21] extended the approach proposed by Wu and Matsumoto [22] by considering the effect of the stress triaxiality in the strain at fracture. According to them, the residual stresses at the machined surface increases with the fracture strain of the work material, since more energy is required to separate the work material to form the chip and to generate the machined surface.

Table 3 summarizes the values of the stress triaxiality, strain at fracture, and temperature ( $T_s$ ) at the zone of the material separation (adjacent to the cutting edge), for the simulated cutting conditions (S1, S2, S3, and S4). Table 3 shows that the changes in the cutting conditions induce a small variation in the temperature  $T_s$ , thus a negligible effect in the material flow stress (less than 1%). Therefore, the residual stresses induced by the cutting conditions above mentioned are due to the mechanical action of the cutting tool on the workpiece only.

Table 3. Summary of the stress triaxiality, strain at fracture, and temperature of the machined surface ( $T_s$ ).

| Simulation ID | $T_s$ [°C] | Triaxiality | Strain at fracture |
|---------------|------------|-------------|--------------------|
| S1            | 280        | 0.77        | 0.19               |
| S2            | 256        | 0.88        | 0.17               |
| S3            | 261        | 0.62        | 0.35               |
| S4            | 235        | 0.70        | 0.23               |

##### 4.2.2. Influence of the cutting speed

Fig. 8 shows that when the cutting speed increases from 30 to 90 m/min, the residual stress become less tensile near the surface and more compressive below it, reaching a maximum in compression of about 650 MPa. Similar results were obtained by Outeiro et al. [23] in machining AISI 316L stainless steel. Table 3 shows that the increase in the cutting speed (from S1 to S2 in Table 3) causes an increase in the stress triaxiality. Thus, the strain to fracture will slightly decrease, and consequently, the residual stress becomes less tensile near the surface.

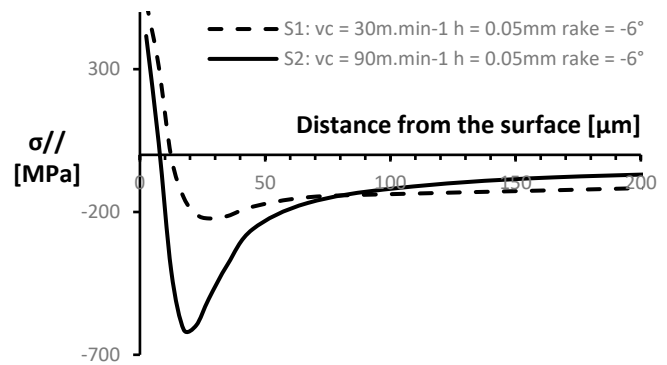


Fig. 8. Influence of the cutting speed on predicted residual stress in the cutting direction ( $w = 4$  mm,  $\alpha = 7^\circ$ ,  $r_n = 25$   $\mu$ m).

##### 4.2.3. Influence of the uncut chip thickness

Fig. 9 shows an increase in the residual stress when the uncut chip thickness increases from 0.05 mm to 0.25 mm. This is due to the decrease in the stress triaxiality with the increase of the uncut chip thickness (from S2 to S3 in Table 3). Thus, the strain to fracture will increase, and consequently, the residual stress.

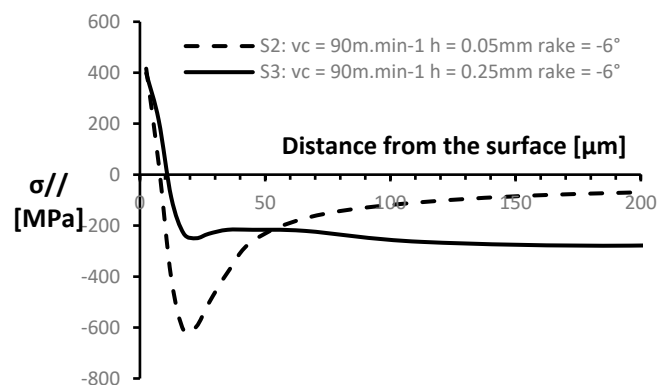


Fig. 9. Influence of the uncut chip thickness on predicted residual stress in the cutting direction. ( $w = 4$  mm,  $\alpha = 7^\circ$ ,  $r_n = 25$   $\mu$ m).

#### 4.2.4. Influence of the tool rake angle

Fig. 10 shows an increase in the residual stress when the tool rake angle changes from a positive value of  $10^\circ$  to a negative value of  $-6^\circ$  generates. This is due to the decrease in the stress triaxiality when the tool rake angle changes from positive to negative (from S4 to S3 in Table 3). Thus, the strain to fracture will increase, and consequently, the residual stress.

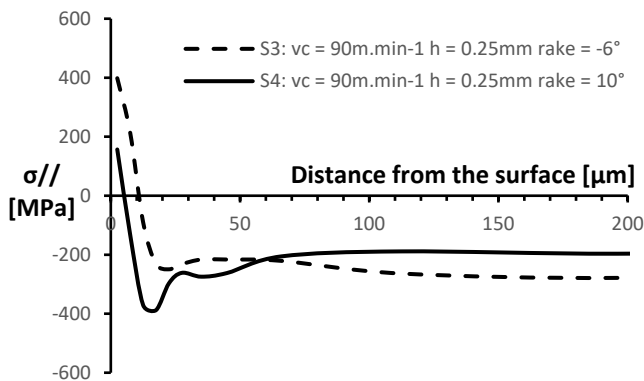


Fig. 10. Influence of the tool rake angle on predicted residual stress in the cutting direction ( $w = 4$  mm,  $\alpha = 7^\circ$ ,  $r_n = 25$   $\mu$ m).

## 5. Conclusions

A constitutive model (plasticity and damage) representing the mechanical behavior of the Inconel 718 nickel-based superalloy in machining is proposed. This model includes the effects of strain rate, temperature, and state of stress (stress triaxiality and lode angle) on the material plasticity and fracture. This model was integrated into an orthogonal cutting model of Inconel 718 nickel-based superalloy to predict the cutting force, chip geometry, cutting temperature, and residual stresses. The comparison between predicted and measured results permitted to conclude that the orthogonal cutting model can represent the cutting process of Inconel 718 with enough accuracy. Therefore, this model was applied to investigate the influence of the cutting speed, uncut chip thickness, and rake angle on the in-depth residual stress profiles. The obtained results show that the residual stresses in the machined surface and subsurface decrease when: a) the cutting speed increases, ii) the uncut chip thickness decrease, and iii) the rake angle increase. Moreover, the lowest cutting speed was the condition that induced the highest tensile residual stress on the surface of the Inconel 718.

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